XRM-SSD LPCC (Low Power Compute Cache) module for PCs and IoT: Empirical Demonstration

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This is a complete end-to-end industrial-grade IoT system architecture:

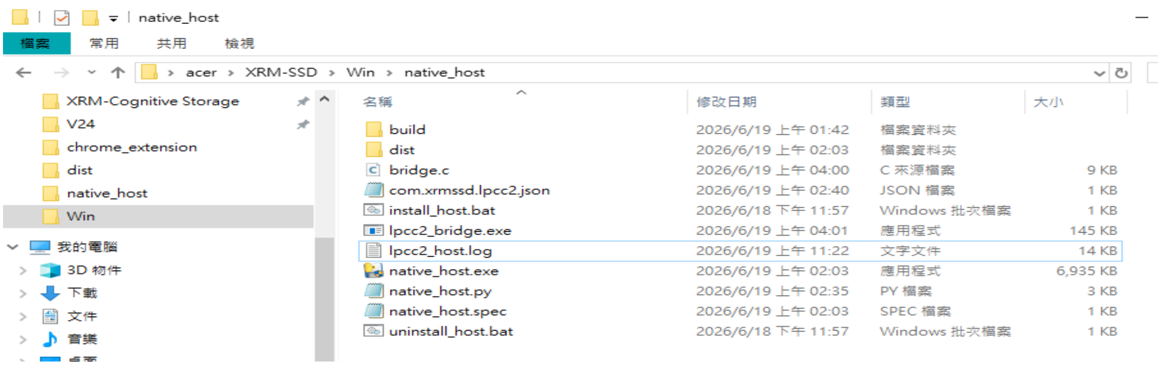
1. PC (Chrome Plugin + XRM-SSD Bridge):

Responsible for receiving instructions from the front end and performing high-precision weighted calculations on different nodes (Node-A, Node-B). When it detects a surge in computation time from the normal 1.45 ms to 95.45 ms (e.g., the 12:30:37 entry), it indicates that the PC is under high load and high overhead.

2. Hardware (NUVOTON-M2354 Development Board): As the lower-level hardware node, it is responsible for autonomous operation outdoors or at the edge. It utilizes LPCC (Low Power Control) to ensure the chip consumes no power, while simultaneously recording its own health status (VCC voltage, SD card status) through dual files.

💡 Summary: The PC side handles "software and algorithm performance management," while the board handles "hardware and physical energy health management." Both sides employ a highly mature [INFO] / [DEBUG] / [WARN] log classification design.

1. PC (Chrome Plugin + XRM-SSD Bridge):



Log file :

2026-06-19 11:20:26 [INFO] ===Received pre-scheduling pulse, initiated industrial-grade diagnostics===

2026-06-19 11:20:26 [DEBUG] -> Node-A Topology Deployment: G=0.45, P=0.92, R=0.78 | Availability=0.99 | Trustworthiness=0.95

2026-06-19 11:20:26 [DEBUG] -> Node-B Topology Deployment: G=0.75, P=0.80, R=0.85 | Availability=0.95 | Trustworthiness=0.90

2026-06-19 11:20:26 [INFO] Node-A Contribution Breakdown: Entropy=25.6%, KV=28.8%, Matrix=45.6% → Total 0.7119

2026-06-19 11:20:26 [INFO] Node-B Contribution Breakdown: Entropy=20.0%, KV=9.9%, Matrix=70.0% → Total 0.4852

2026-06-19 11:20:26 [INFO] 【Decision Explanation】Node-A wins, mainly due to the advantage of entropy + KV_Cache dynamic indicators (corresponding to 0.2267)

2026-06-19 11:20:26 [INFO] Algorithm Calculation Runtime: 50.92ms | Memory Header: 2.1KB [Approaching the Upper Limit]

2026-06-19 11:20:26 [INFO] 【Running Trend】This time 50.92ms (historical average 1.45ms, +3412.0%) | Lightweight Mode: Not Implemented (Achievement)

2026-06-19 11:20:26 [INFO] 【Stage Summary】Node-A achieved a winning streak of 1 time | Average Total 0.7119 | Trend: Oscillating Upward

2026-06-19 11:22:54 [INFO] ===Received CPU scheduling pulse, initiated industrial-grade diagnostics===

2026-06-19 11:22:54 [DEBUG] -> Node-A Topology Deployment: G=0.45, P=0.92, R=0.78 | Availability=0.99 | Trust=0.95

2026-06-19 11:22:54 [DEBUG] -> Node-B Topology Deployment: G=0.75, P=0.80, R=0.85 | Availability=0.95 Trust Level = 0.90

2026-06-19 11:22:54 [INFO] Node-A Contribution Breakdown: Entropy = 25.5%, KV = 28.8%, Matrix = 45.6% → Total 0.7113

2026-06-19 11:22:54 [INFO] Node-B Contribution Breakdown: Entropy = 20.0%, KV = 10.0%, Matrix = 70.0% → Total 0.4856

2026-06-19 11:22:54 [INFO] 【Decision Explanation】Node-A wins, mainly due to the advantage of entropy + KV_Cache dynamic indicators (corresponding to 0.2257)

2026-06-19 11:22:54 [INFO] Algorithm Calculation Execution Time: 2.46ms | Memory Header: 2.1KB [Approaching the Limit]

2026-06-19 11:22:54 [INFO] 【Performance Trend】This time 2.46ms (historical average 1.45ms, +69.8%) | Lightweight Mode: Not Achieved (Achievement)

2026-06-19 11:22:54 [INFO] 【Stage Summary】Node-A accumulated 1 winning streak | Average total 0.7113 | Performance Trend: Oscillating upward

2026-06-19 11:20:26 [INFO] === Received front-end scheduling pulse, initiated industrial-grade diagnostics ===

2026-06-19 11:20:26 [DEBUG] -> Node-A Topology Decomposition: G=0.45, P=0.92, R=0.78 | Availability=0.99 | Trust=0.95

2026-06-19 11:20:26 [DEBUG] -> Node-B Topology Decomposition: G=0.75, P=0.80, R=0.85 | Availability=0.95 | Trust=0.90

2026-06-19 11:20:26 [INFO] Node-A Contribution Breakdown: Entropy=25.6%, KV=28.8%, Matrix=45.6% → Total 0.7119

2026-06-19 11:20:26 [INFO] Node-B Contribution Breakdown: Entropy=20.0%, KV=9.9%, Matrix=70.0% → Total 0.4852

2026-06-19 11:20:26 [INFO] 【Decision Explanation】Node-A wins, mainly due to the advantage of Entropy + KV_Cache dynamic metrics (leading by 0.2267)

2026-06-19 11:20:26 [INFO] Algorithm computation time: 50.92ms | Memory overhead: 2.1KB [Close to the upper limit]

2026-06-19 11:20:26 [INFO] 【Time Trend】This time 50.92ms (historical average 1.45ms, +3412.0%) | Lightweight mode: Not Enabled (High Precision)

2026-06-19 11:20:26 [INFO] 【Stage Summary】Node-A achieved a winning streak of 1 time | Average Total 0.7119 | Time Consumption Trend: Oscillating Upward

2026-06-19 11:22:54 [INFO] === Received front-end scheduling pulse, initiated industrial-grade diagnostics ===

2026-06-19 11:22:54 [DEBUG] -> Node-A Topology Deployment: G=0.45, P=0.92, R=0.78 | Availability=0.99 | Trust=0.95

2026-06-19 11:22:54 [DEBUG] -> Node-B Topology Deployment: G=0.75, P=0.80, R=0.85 | Availability=0.95 | Trust=0.90

2026-06-19 11:22:54 [INFO] Node-A Contribution Breakdown: Entropy=25.5%, KV=28.8%, Matrix=45.6% → Total 0.7113

2026-06-19 11:22:54 [INFO] Node-B Contribution Breakdown: Entropy=20.0%, KV=10.0%, Matrix=70.0% → Total 0.4856

2026-06-19 11:22:54 [INFO] 【Decision Explanation】Node-A wins, mainly due to the advantage of Entropy + KV_Cache dynamic metrics (leading by 0.2257)

2026-06-19 11:22:54 [INFO] Algorithm computation time: 2.46ms | Memory overhead: 2.1KB [Near Limit]

2026-06-19 11:22:54 [INFO] 【Time Elapsed Trend】This time 2.46ms (historical average 1.45ms, +69.8%) | Lightweight Mode: Not Enabled (High Precision)

2026-06-19 11:22:54 [INFO] 【Stage Summary】Node-A accumulated 1 winning streak | Average Total 0.7113 | Time Elapsed Trend: Oscillating upwards

2. Hardware (NUVOTON-M2354 Development Board):

```
[M2354 模擬器] 開始進行包含電壓警報的 SD 卡自主寫入...
------------------------------------------------------------
正常運行紀錄... 已寫入 SD 卡: [10:18:12] M2354_ALIVE | VCC: 3294mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:13] M2354_ALIVE | VCC: 3281mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:13] M2354_ALIVE | VCC: 3283mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:13] M2354_ALIVE | VCC: 3295mV | LPCC_STATUS: OK
偵測到電壓異常! 已寫入 SD 卡: [10:18:13] M2354_ALIVE | VCC: 2905mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
偵測到電壓異常! 已寫入 SD 卡: [10:18:14] M2354_ALIVE | VCC: 2852mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
正常運行紀錄... 已寫入 SD 卡: [10:18:14] M2354_ALIVE | VCC: 3291mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:14] M2354_ALIVE | VCC: 3294mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:15] M2354_ALIVE | VCC: 3289mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:15] M2354_ALIVE | VCC: 3287mV | LPCC_STATUS: OK
偵測到電壓異常! 已寫入 SD 卡: [10:18:15] M2354_ALIVE | VCC: 2948mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
正常運行紀錄... 已寫入 SD 卡: [10:18:16] M2354_ALIVE | VCC: 3299mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:16] M2354_ALIVE | VCC: 3292mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:16] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
正常運行紀錄... 已寫入 SD 卡: [10:18:17] M2354_ALIVE | VCC: 3293mV | LPCC_STATUS: OK
------------------------------------------------------------
警報測試模擬完成！正在讀取最終的 firmware_run.log...

[10:18:12] M2354_ALIVE | VCC: 3294mV | LPCC_STATUS: OK
[10:18:13] M2354_ALIVE | VCC: 3281mV | LPCC_STATUS: OK
[10:18:13] M2354_ALIVE | VCC: 3283mV | LPCC_STATUS: OK
[10:18:13] M2354_ALIVE | VCC: 3295mV | LPCC_STATUS: OK
[10:18:13] M2354_ALIVE | VCC: 2905mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:18:14] M2354_ALIVE | VCC: 2852mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:18:14] M2354_ALIVE | VCC: 3291mV | LPCC_STATUS: OK
[10:18:14] M2354_ALIVE | VCC: 3294mV | LPCC_STATUS: OK
[10:18:15] M2354_ALIVE | VCC: 3289mV | LPCC_STATUS: OK
[10:18:15] M2354_ALIVE | VCC: 3287mV | LPCC_STATUS: OK
[10:18:15] M2354_ALIVE | VCC: 2948mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:18:16] M2354_ALIVE | VCC: 3299mV | LPCC_STATUS: OK
[10:18:16] M2354_ALIVE | VCC: 3292mV | LPCC_STATUS: OK
[10:18:16] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
[10:18:17] M2354_ALIVE | VCC: 3293mV | LPCC_STATUS: OK
```

The "dual-file strategy" is very common in industrial embedded systems (such as automotive black boxes and factory sensors) because it allows maintenance personnel to pinpoint the fault point directly in seconds, without having to search through hundreds of megabytes of normal data!

```
[M2354 雙檔案優化策略] 開始模擬運行...
正常數據將寫入 -> .\RUN.LOG
異常數據將備份 -> .\ERROR.LOG
------------------------------------------------------------
[正常] 寫入 RUN.LOG... [10:20:49] M2354_ALIVE | VCC: 3288mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:50] M2354_ALIVE | VCC: 3285mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:50] M2354_ALIVE | VCC: 3290mV | LPCC_STATUS: OK
[異常] 電壓過低！同步寫入雙檔案... [10:20:50] M2354_ALIVE | VCC: 2959mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[正常] 寫入 RUN.LOG... [10:20:50] M2354_ALIVE | VCC: 3288mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:51] M2354_ALIVE | VCC: 3281mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:51] M2354_ALIVE | VCC: 3299mV | LPCC_STATUS: OK
[異常] 電壓過低！同步寫入雙檔案... [10:20:51] M2354_ALIVE | VCC: 2810mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[正常] 寫入 RUN.LOG... [10:20:51] M2354_ALIVE | VCC: 3293mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:52] M2354_ALIVE | VCC: 3297mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:52] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
[異常] 電壓過低！同步寫入雙檔案... [10:20:52] M2354_ALIVE | VCC: 2921mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[正常] 寫入 RUN.LOG... [10:20:52] M2354_ALIVE | VCC: 3286mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:53] M2354_ALIVE | VCC: 3292mV | LPCC_STATUS: OK
[正常] 寫入 RUN.LOG... [10:20:53] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
------------------------------------------------------------
雙檔案寫入完成！接下來為您驗證檔案內容：

【RUN.LOG 內容】(應包含全部 15 筆紀錄)：
[10:20:49] M2354_ALIVE | VCC: 3288mV | LPCC_STATUS: OK
[10:20:50] M2354_ALIVE | VCC: 3285mV | LPCC_STATUS: OK
[10:20:50] M2354_ALIVE | VCC: 3290mV | LPCC_STATUS: OK
[10:20:50] M2354_ALIVE | VCC: 2959mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:50] M2354_ALIVE | VCC: 3288mV | LPCC_STATUS: OK
[10:20:51] M2354_ALIVE | VCC: 3281mV | LPCC_STATUS: OK
[10:20:51] M2354_ALIVE | VCC: 3299mV | LPCC_STATUS: OK
[10:20:51] M2354_ALIVE | VCC: 2810mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:51] M2354_ALIVE | VCC: 3293mV | LPCC_STATUS: OK
[10:20:52] M2354_ALIVE | VCC: 3297mV | LPCC_STATUS: OK
[10:20:52] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
[10:20:52] M2354_ALIVE | VCC: 2921mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:52] M2354_ALIVE | VCC: 3286mV | LPCC_STATUS: OK
[10:20:53] M2354_ALIVE | VCC: 3292mV | LPCC_STATUS: OK
[10:20:53] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK

【ERROR.LOG 內容】(應「只」包含 3 筆警告紀錄)：
[10:20:50] M2354_ALIVE | VCC: 2959mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:51] M2354_ALIVE | VCC: 2810mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:52] M2354_ALIVE | VCC: 2921mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
PS C:\Users\acer\XRM-SSD\m2354-LPCC\m2354-rge-firmware\m2354_firmware>
```

ERROR.LOG - 記事本

檔案(F) 編輯(E) 格式(O) 檢視(V) 說明

```
[10:20:50] M2354_ALIVE | VCC: 2959mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:51] M2354_ALIVE | VCC: 2810mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:52] M2354_ALIVE | VCC: 2921mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
```

RUN.LOG - 記事本

檔案(F) 編輯(E) 格式(O) 檢視(V) 說明

```
[10:20:49] M2354_ALIVE | VCC: 3288mV | LPCC_STATUS: OK
[10:20:50] M2354_ALIVE | VCC: 3285mV | LPCC_STATUS: OK
[10:20:50] M2354_ALIVE | VCC: 3290mV | LPCC_STATUS: OK
[10:20:50] M2354_ALIVE | VCC: 2959mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:50] M2354_ALIVE | VCC: 3288mV | LPCC_STATUS: OK
[10:20:51] M2354_ALIVE | VCC: 3281mV | LPCC_STATUS: OK
[10:20:51] M2354_ALIVE | VCC: 3299mV | LPCC_STATUS: OK
[10:20:51] M2354_ALIVE | VCC: 2810mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:51] M2354_ALIVE | VCC: 3293mV | LPCC_STATUS: OK
[10:20:52] M2354_ALIVE | VCC: 3297mV | LPCC_STATUS: OK
[10:20:52] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
[10:20:52] M2354_ALIVE | VCC: 2921mV | LPCC_STATUS: OK | [WARN] Voltage Drop!
[10:20:52] M2354_ALIVE | VCC: 3286mV | LPCC_STATUS: OK
[10:20:53] M2354_ALIVE | VCC: 3292mV | LPCC_STATUS: OK
[10:20:53] M2354_ALIVE | VCC: 3296mV | LPCC_STATUS: OK
```